



NATIONAL SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2023

MATHEMATICAL LITERACY P2 MARKING GUIDELINE

MARKS: 100

Symbol	Explanation
M	Method
MA	Method with accuracy
CA	Consistent accuracy
RCA	Rounding consistent accuracy
A	Accuracy
C	Conversion
S	Simplification
SF	Correct substitution in a formula
J	Justification
O	Opinion/Example/Definition/Explanation/Justification/Verification
RT/RG/RM	Reading from a table/graph/map
P	Penalty, e.g. for no units, incorrect rounding off etc.
R	Rounding off
NPR	No penalty rounding or omitting units
AO	Answer only, full marks

This marking guideline consists of 9 pages.

MARKING GUIDELINES**NOTE:**

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled version).
- Consistent Accuracy (CA) applies in ALL aspects of the marking guidelines; however, it stops at the second calculation error.
- If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalise for every extra incorrect item presented.

LET WEL:

- *As 'n kandidaat 'n vraag TWEE keer beantwoord merk slegs die EERSTE poging.*
- *As 'n kandidaat 'n antwoord van 'n vraag doodtrek (kanselleer) en nie oordoen nie, merk die doodgetrekte (gekanselleerde) poging.*
- *Volgehoue akkuraatheid (CA) word in ALLE aspekte van die nasienriglyn toegepas, maar dit hou by die tweede berekeningsfout op.*
- *Wanneer 'n kandidaat aflees van 'n grafiek, tabel, uitlegplan en kaart en ekstra antwoorde gee, penaliseer vir elke ekstra item.*

KEY TO TOPIC SYMBOL:**F = Finance; M = Measurement; MP = Maps, plans and other representations; P= Probability****QUESTION 1 [23]**

Quest	Solution	Explanation	Level
1.1.1	$3,142 \times 450 \checkmark M$ $= 1\,413,9 \text{ mm} \checkmark A$	1M multiply by 450 1A correct answer (2)	M L1
1.1.2	Option B: $290 \div 200 \text{ m} \checkmark M$ $= R1,45 \text{ per metre} \checkmark MA$	1M divide by 200 m 1MA answer (2)	M L1
1.1.3	$R290 : R390 \checkmark M$ $29 : 39 \checkmark A$	1M correct order 1A correct simplified values (2)	M L1
1.1.4	$\text{Radius} = 450 \text{ mm} \div 10 \checkmark C$ $= 45 \text{ cm} \div 2 \checkmark M$ $= 22,5 \text{ cm} \checkmark A$	1C divide by 10 1M finding the radius 1A correct answer (3)	M L1
1.2.1	$N2; N3; N6; N10 \checkmark \checkmark A$ (Any TWO)	2A correct national roads (2)	MP L1
1.2.2	Strip Chart / Map $\checkmark \checkmark A$	2A name (2)	MP L1
1.2.3	$\checkmark RT \quad \checkmark C$ $\text{Distance} = 964 \text{ km} \times 1000$ $= 964\,000 \text{ m} \checkmark A$	1RT correct values 1C conversion 1A 964 000 m (3)	MP L1

1.2.4 (a)	Port St. Johns OR Port Edward ✓ ✓ A	2 A town (2)	MP L1
1.2.4 (b)	✓ A Distance = 201 + 36 ✓ M = 237 km ✓ A	1A correct distances 1M Addition 1A 237 km (3)	MP L1
1.2.5	R617; R101; R56 ; R61 ✓ ✓ RM (Accept any two answers)	2RM correct provincial roads (2)	MP L1
		[23]	

QUESTION 2 [18]			
Quest.	Solution	Explanation	Level
2.1.1	Bar Scale ✓✓ RM	2RM correct answer (2)	MP L1
2.1.2	South West ✓✓ RM	2RM correct direction (2)	MP L2
2.1.3	✓M 2 cm : 250 km ✓M 9,6 cm : $\frac{9,6 \times 250}{2}$ $\approx 1\,200$ km ✓ CA Mr. Antonie is incorrect ✓ J	1M measured value 1M multiply by 250 and divide by 2 1CA correct answer 1J correct justification (4)	MP L3
2.2.1	Average Speed = $\frac{1\,635}{17,25}$ ✓ MA $= 94,782$ ✓ A $= 94,78$ km / hr ✓ R	1MA divide by 17,25 1A correct answer 1R correct rounding (3)	M L2
2.2.2 (a)	1 litre = 12,5 km No. of litres = $\frac{1\,635}{12,5} \times 1$ ✓ MA $= 130,8$ ✓ A 0,80 litre = 10 km No of litres = $\frac{1\,635}{10} \times 0,80$ ✓ MA $= 130,8$ ✓ A Mr Amos statement is incorrect. ✓ J	1MA dividing by 12,5 1A correct answer 1MA dividing by 10 1A correct answer 1J correct deduction (5)	MP L4
2.2.2 (b)	Cost = $130,8 \times R24,75$ ✓ M $= R3\,237,30$ ✓ A	1M multiply by 130,8 1A correct answer (2)	M L2
		[18]	

QUESTION 3 [32]			
Quest	Solution	Explanation	Level
3.1.1	Mass of cake = $900 \div 1\,000$ ✓M $= 0,9 \text{ kg}$ ✓A	1M divide by 1000 1A correct answer (2)	M L1
3.1.2	Mass of one slice = $\frac{900}{12}$ ✓ M $= 75 \text{ g}$ ✓ A	1M divide by 12 1A correct answer (2)	M L2
3.1.3	Calories of one slice of cake = $\frac{75}{100} \times 400$ ✓✓ M $= 300 \text{ calories}$ ✓A	1M multiply by 75 1M divide by 100 1A correct answer (3)	M L2
3.1.4	Convert min to hrs ✓M $75 \text{ min} \div 60 = 1,25 \text{ hrs}$ ✓ A	1M conversion ratio 1A correct answer (2)	M L1
3.1.5	$90 \text{ guests} = 90 \text{ slices}$ ✓M Number of Cakes = $90 \div 12$ ✓M $= 7,5 \text{ cakes}$ ✓CA $\approx 8 \text{ cakes}$ ✓R	1M number of slices 1M divide by 12 1CA number of cakes 1A correct rounding (4)	M L1
3.2.1	$\text{Cups} = 8 \times \frac{3}{4}$ ✓M $= 6 \text{ cups}$ ✓A	1M multiplication 1A correct answer (2)	M L2
3.2.2	$\text{Cocoa} = 8 \times 90 \text{ g}$ ✓M Required Amount = $720 \text{ g} \div 240 \text{ g}$ $= 3$ ✓M ✓M Cost = $3 \times \text{R } 62,75$ $= \text{R}188,25$ ✓A	1M total grams 1M amount 1M multiplication 1A correct cost (4)	M L3

3.2.3	$^{\circ}\text{C} = (^{\circ}\text{F} - 32) \div 1,8$ $\checkmark\text{SF}$ $= (320 - 32) \div 1,8$ $\checkmark\text{M}$ $= 288 \div 1,8$ $= 160^{\circ}\text{C} \checkmark\text{CA}$	1SF substitution 1M $268 \div 1,8$ 1CA correct rounding (3)	M L2
3.2.4	Starting time 09h 03 + 55 min $\checkmark\text{M}$ Finishing time = 09h 58 $\checkmark\text{A}$	1M adding time 1A correct answer (2)	M L2
3.3.1	Radius = $86 \text{ mm} \div 2 = 43 \text{ mm} \checkmark\text{A}$ Convert: $43 \text{ mm} \div 10 = 4,3 \text{ cm} \checkmark\text{C}$ Volume of one can = $3,142 \times 4,3 \times 4,3 \times \text{height} \checkmark\text{SF}$ $546,10 \text{ cm}^3 = 58,09558 \text{ cm}^2 \times \text{height}$ Height = $546,10 \text{ cm}^3 \div 58,09558 \text{ cm}^2$ = $9,4 \text{ cm} \checkmark\text{CA}$	1A radius value 1C conversion 1SF correct values 1CA height value (4)	M L3
3.3.2	$\checkmark\text{M}$ Height of label = $80\% \times 9,4 \text{ cm}$ = $7,52 \text{ cm} \checkmark\text{A}$ Difference = $9,4 - 7,52$ = $1,88 \text{ cm} \checkmark\text{M}$ His statement is invalid $\checkmark\text{J}$	CA value from Q 3.3.1 1M calculating 80% of 9,4 1A for 7,52 cm 1M difference value of 1,88 cm 1J justification (4)	M L3
			[32]

QUESTION 4 [27]			
Quest	Solution	Explanation	Level
4.1.1	Shows a building's plan as seen from above. It is a 2-dimensional view of the building. ✓✓ A	2A correct explanation (2)	MP L1
4.1.2	One window ✓✓ A	2A correct answer (2)	MP L2
4.1.3	1 mm represents 100 mm Length of wall ✓ M $114 \text{ mm} = 114 \times 100 \text{ ✓ M}$ $= 11\,400 \text{ mm} \div 1\,000 \text{ ✓ M}$ $= 11,4 \text{ m ✓ CA}$ Width of Wall ✓ M $77 \text{ mm} = 77 \times 100$ $= 7\,700 \div 1\,000 \text{ ✓ M}$ $= 7,7 \text{ m ✓ CA}$	1M for measurement 1MA using conversion factor 1M conversion to m 1CA for 11,4 m 1M for measurement 1M conversion to m 1CA 7,7 m (7)	MP L4
4.1.4	✓ MA $\text{Width} = 19,38 \text{ m}^2 \div 10,2 \text{ m}$ $= 1,9 \text{ m ✓ A}$ Times lesser = $10,2 \div 1,9 \text{ m ✓ MA}$ $= 5,37 \text{ times}$ Mrs Smith's statement is invalid. ✓ J	1MA $19,38 \div 10,2$ 1A correct answer 1MA divide by correct values 1J correct justification (4)	M L4
4.2.1	Area Circle = $\pi \times r^2$ $= 3,142 \times 1,65 \times 1,65 \text{ ✓ SF}$ $= 8,55 \text{ cm}^2 \text{ ✓ A}$ Area square = side $\times$ side $= 0,9 \times 0,9$ $= 0,81 \text{ cm}^2 \text{ ✓ A}$ Area of coin = $8,55 - 0,81 \text{ ✓ M}$ $= 7,74 \text{ cm}^2$ $= 7,7 \text{ cm}^2 \text{ ✓ A}$	1SF correct radius 1A correct answer 1A correct answer 1M subtraction 1A correct rounded off (5)	M L3
4.2.2	Mass of coin = $1,47 \times 19,30 \text{ ✓ M}$ $= 28,371 \text{ grams}$ $\approx 28,4 \text{ g ✓ A}$	1M correct values 1A correct answer (2)	M L2

